GeoAI Based Land Suitability Assessment and Crop Yield Prediction System

Project Work I — Review

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01 · MOTIVATION

Problem Statement

1

Tradition over data

Crop choices rely on experience, not systematic soil, terrain, or climate evaluation.

2

Static, expert-driven tools

Frameworks like AHP depend on fixed expert weights and don't adapt per field.

3

Coarse resolution

Suitability tools operate at district/block level, missing small and marginal landholders.

4

Low transparency

Existing yield models offer little insight into which factors drive predictions.

5

No integrated system

No accessible tool fuses soil, terrain, and satellite data into one plot-level pipeline.

02 · OVERVIEW

Abstract

A GeoAI-based framework for plot-level land suitability classification and crop yield prediction — piloted on Coimbatore district, Tamil Nadu.

1

Suitability Classifier

Categorizes plots using soil, terrain & environmental data

2

Crop Recommendation

Suggests the most viable crop for a given plot

3

Yield Prediction (LSTM)

Forecasts yield using temporal & spatial patterns

4

SHAP Explainability

Interprets predictions and key driving factors

Combines machine learning with interpretable AI to deliver a scalable, transparent decision-support tool for agricultural planning.

Literature Survey

2023
Scientific Reports

Simultaneous multi-crop land suitability prediction from remote sensing data using semi-supervised learning

MLP predicts suitability for six crops at once from GEE soil/climate/landscape data, downscaling district-level labels to farm level. Cites the closest prior Tamil Nadu work (RF/DT/SVM, single district, suitability only).

2024
Comm. in Soil Science & Plant Analysis

Machine learning based crop suitability prediction, Barmer district, Thar Desert

MLR, Random Forest & XGBoost classify wheat/pearl millet suitability from digital soil mapping. Strongest direct comparison for the suitability stage, though at coarser district-level resolution.

2025
Intl. Journal of Digital Earth

Sustainable crop recommendation using a seasonally adaptive recursive spectral CNN

Tamil Nadu-specific deep learning model recommending crops from humidity, rainfall, sunlight, soil & temperature, handling seasonal variation. Closest regional precedent for crop recommendation.

2023
Applied Artificial Intelligence

Paddy yield prediction in Tamil Nadu delta region using MLR-LSTM model

Hybrid MLR + LSTM predicts paddy yield over time-series data in the TN delta. Closest regional precedent for yield prediction; hybrid design is a fallback if pure LSTM is data-limited at plot level.

Dataset and Tech Stack

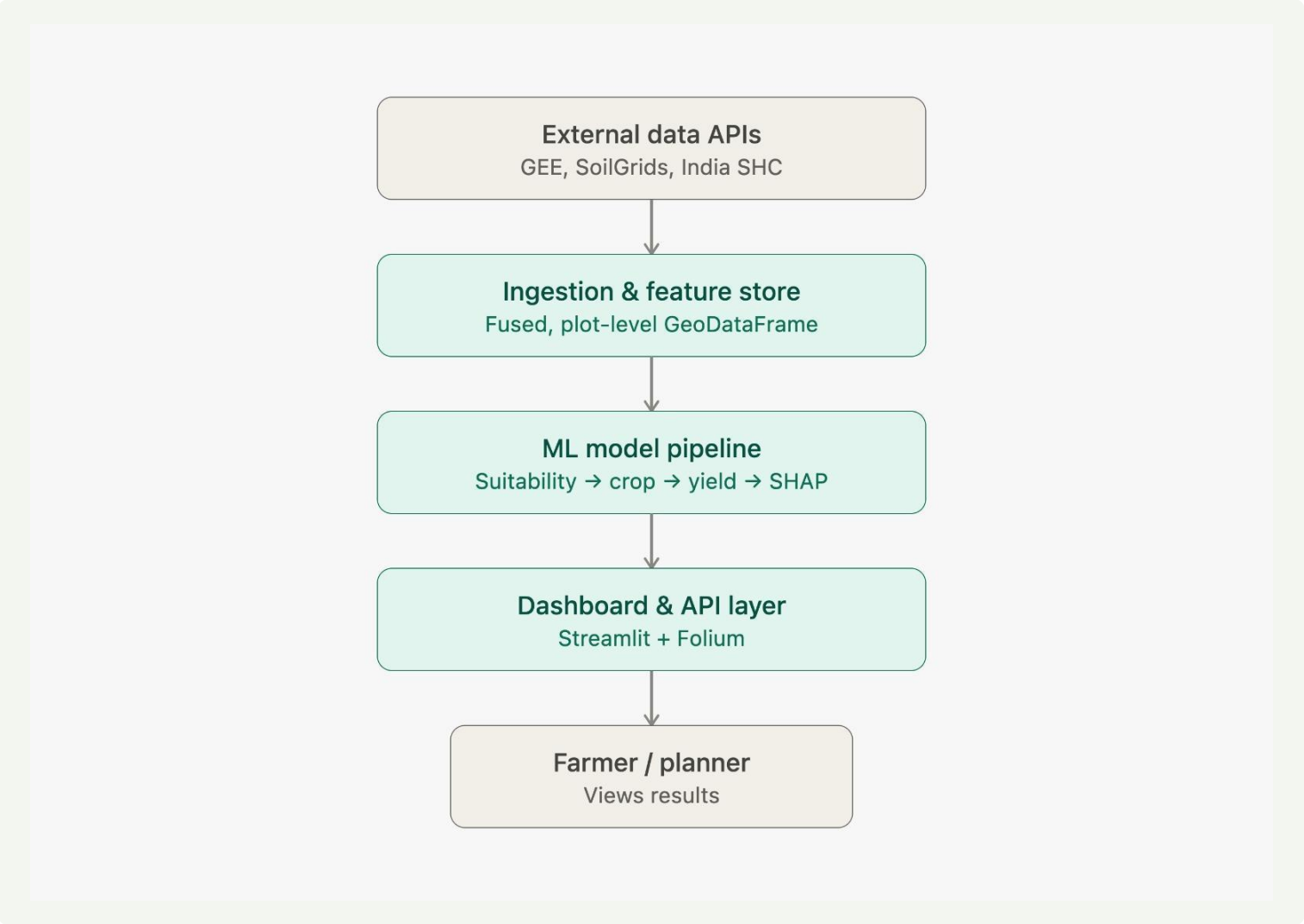
DATASET

- Study area: Coimbatore district, Tamil Nadu
- ~3,725 plots × 10 attributes (1 km grid)
- Terrain — elevation, slope, aspect (SRTM via GEE)
- Vegetation — NDVI (Sentinel-2 via GEE)
- Soil — pH, organic carbon, clay, sand (SoilGrids)
- Boundaries — Tamil Nadu district shapefiles (GitHub)
- Storage — Google Drive (GEE exports)

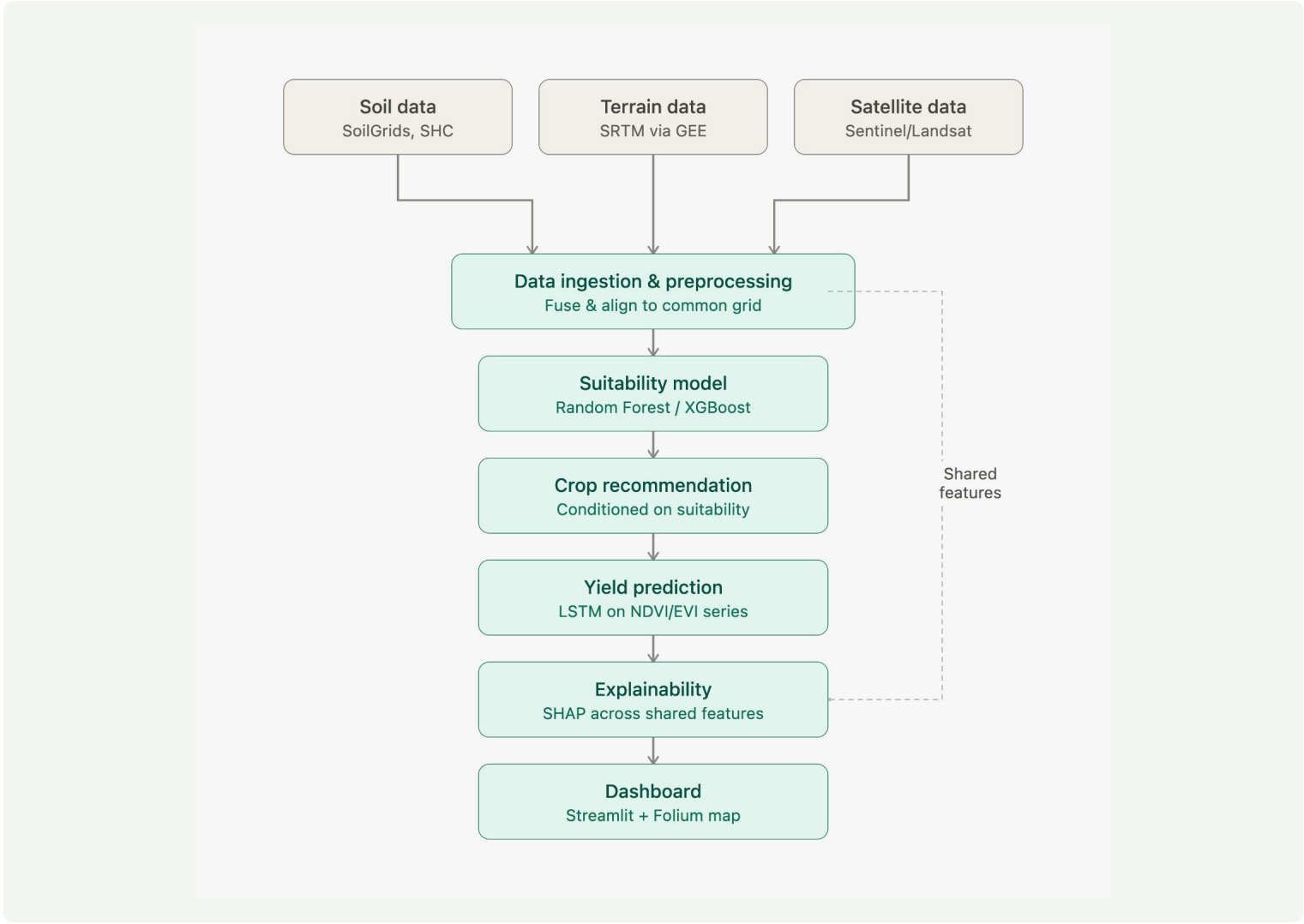
TECH STACK

- Python 3.x · Jupyter / Google Colab
- Rasterio, GeoPandas, Shapely — geospatial processing
- Pandas, NumPy — data handling
- Scikit-learn — suitability & crop recommendation
- TensorFlow / Keras — LSTM yield prediction
- SHAP — model explainability
- Matplotlib, Seaborn — visualization
- Git / GitHub — version control

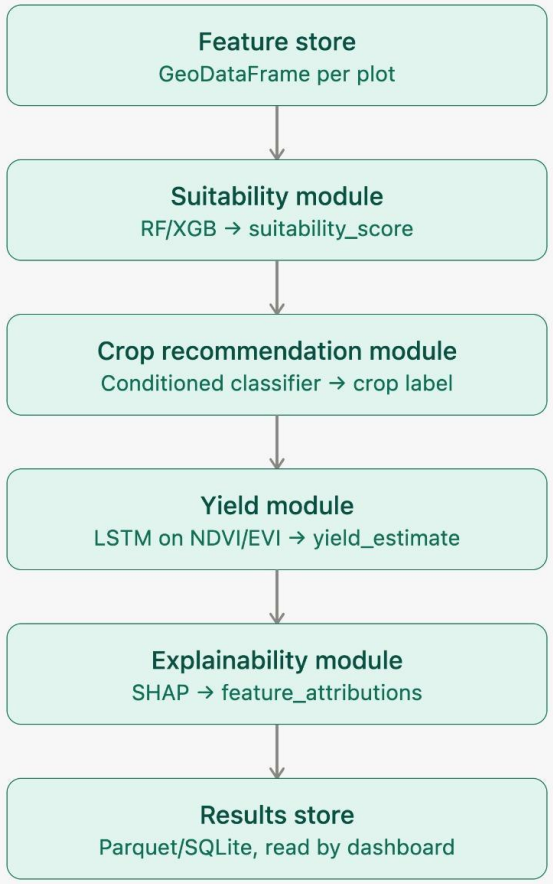
System Architecture



Data Flow Diagram



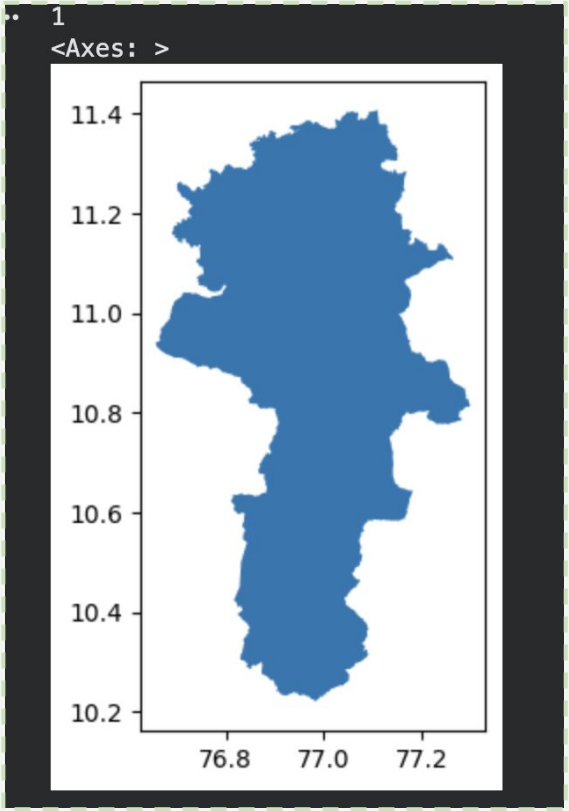
Module / Pipeline Design



Implementation

Step 1.1 — Study Area

```
1
<Axes: >
```



Step 1.2 — Plot Grid

```
print(len(plot_grid))
```

```
3902
```

Implementation

Step 1.4 — Terrain Data

```
Mounted at /content/drive
(3902, 5)
```

	system:index	aspect	elevation	slope	.geo
0	0_0	168.94258	791	23.211750	{"geodesic":false,"type":"Point","coordinates"...
1	1_0	219.54485	842	18.566063	{"geodesic":false,"type":"Point","coordinates"...
2	2_0	111.48539	1153	5.051475	{"geodesic":false,"type":"Point","coordinates"...
3	3_0	106.25545	631	24.827124	{"geodesic":false,"type":"Point","coordinates"...
4	4_0	34.11599	729	6.691145	{"geodesic":false,"type":"Point","coordinates"...

Implementation

Step 1.5 —Vegetation Data

(3902, 3)

	system:index	NDVI	.geo
0	0_0	0.821879	{"geodesic":false,"type":"Point","coordinates"...
1	1_0	0.791460	{"geodesic":false,"type":"Point","coordinates"...
2	2_0	0.717585	{"geodesic":false,"type":"Point","coordinates"...
3	3_0	0.841128	{"geodesic":false,"type":"Point","coordinates"...
4	4_0	0.714020	{"geodesic":false,"type":"Point","coordinates"...

Implementation

Step 1.6 — Soil Data

(3902, 6)

	phh2o	soc	clay	sand	lon	lat
0	53	509	344	376	76.925797	10.24012
1	54	549	335	374	76.915797	10.24012
2	54	406	365	334	76.925797	10.25012
3	53	451	353	381	76.935797	10.24012
4	53	373	349	379	76.945797	10.24012

Implementation

Step 1.7– Final Dataset

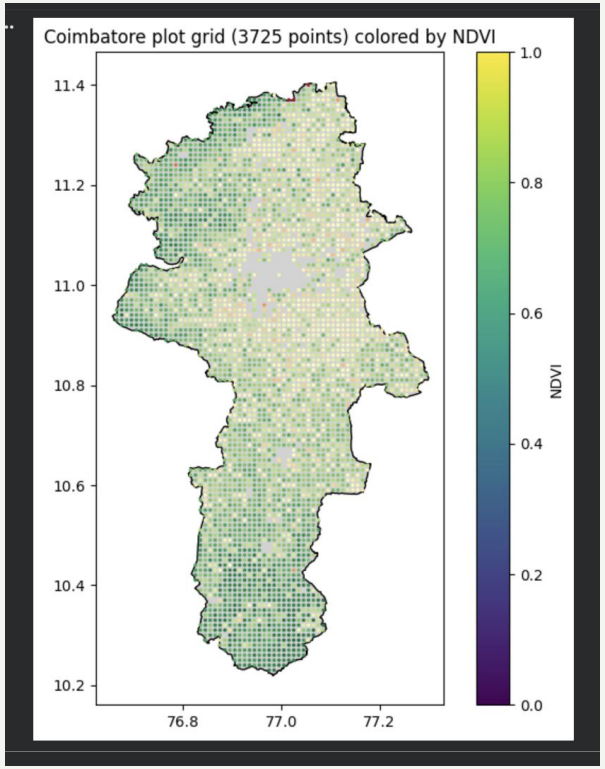
	elevation	slope	aspect	NDVI	phh2o	soc	clay	sand	lon	lat
0	791	23.211750	168.942580	0.821879	5.3	509	34.4	37.6	76.925797	10.24012
1	842	18.566063	219.544850	0.791460	5.4	549	33.5	37.4	76.915797	10.24012
2	1153	5.051475	111.485390	0.717585	5.4	406	36.5	33.4	76.925797	10.25012
3	631	24.827124	106.255450	0.841128	5.3	451	35.3	38.1	76.935797	10.24012
4	729	6.691145	34.115990	0.714020	5.3	373	34.9	37.9	76.945797	10.24012
5	1067	13.283790	123.265940	0.793107	5.2	590	35.6	34.2	76.935797	10.25012
6	945	53.490837	114.041560	0.769461	5.5	576	35.3	35.6	76.945797	10.25012
7	1094	24.449772	102.334640	0.861287	5.3	463	33.7	37.6	76.945797	10.26012
8	1197	7.437497	7.239587	0.735688	5.4	378	33.5	36.0	76.935797	10.26012
9	1084	7.123535	247.133320	0.766200	5.6	432	34.3	33.5	76.915797	10.25012

Implementation

Step 1.8– Data Cleaning

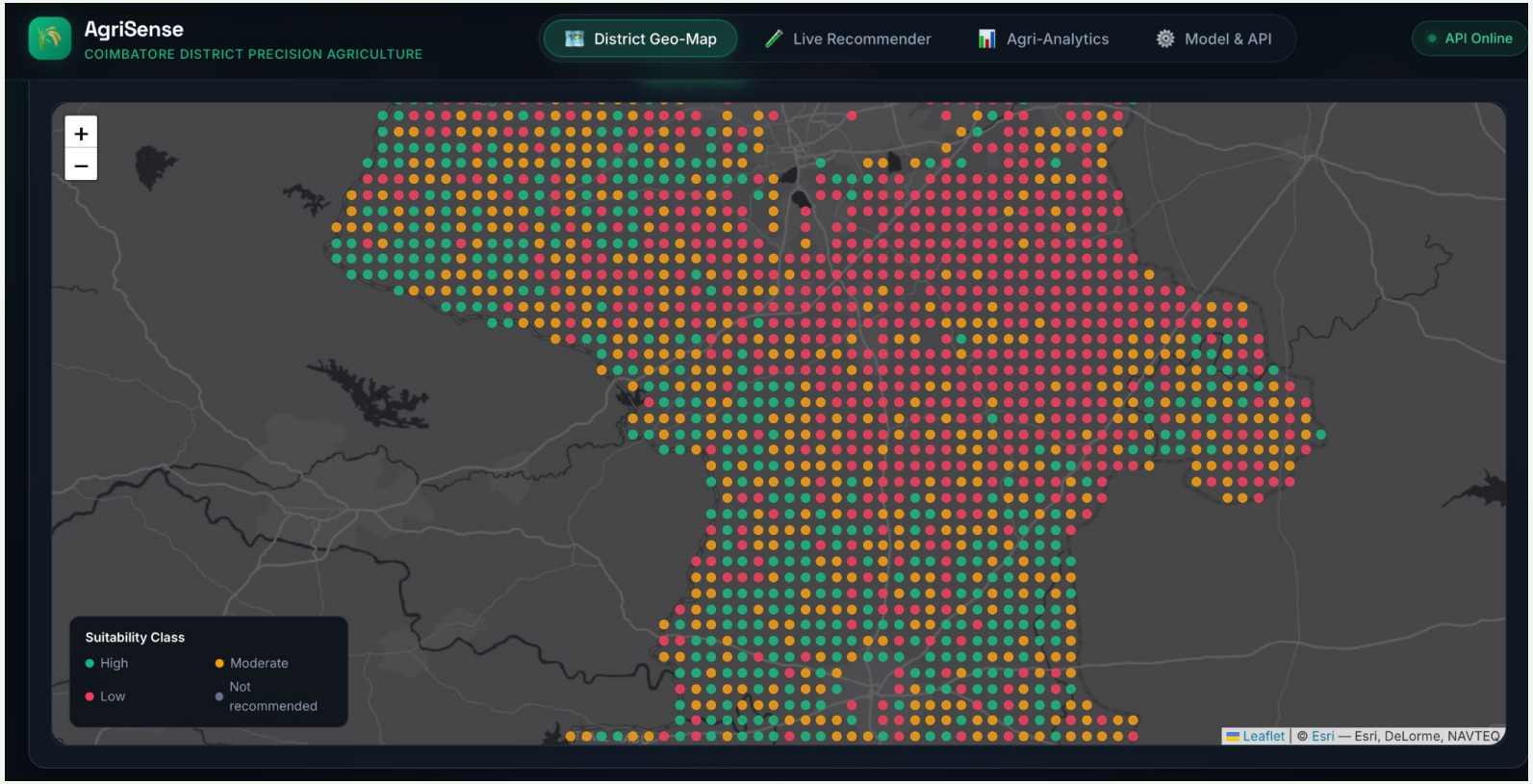
```
... Rows with no-data sentinel values: 176
(3725, 10)
```

Step 1.10 Visualization



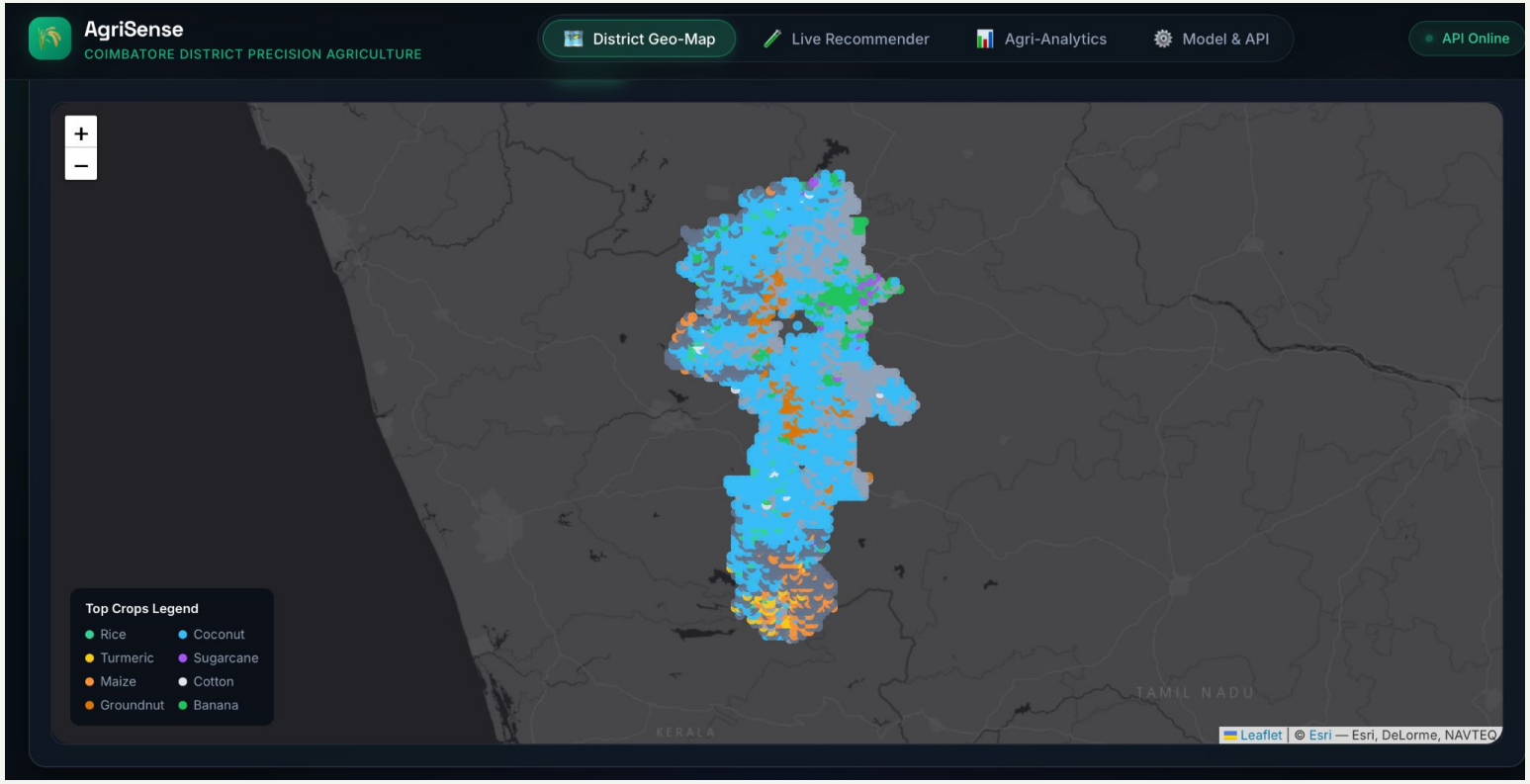
Implementation

Step 2 - Suitability Model



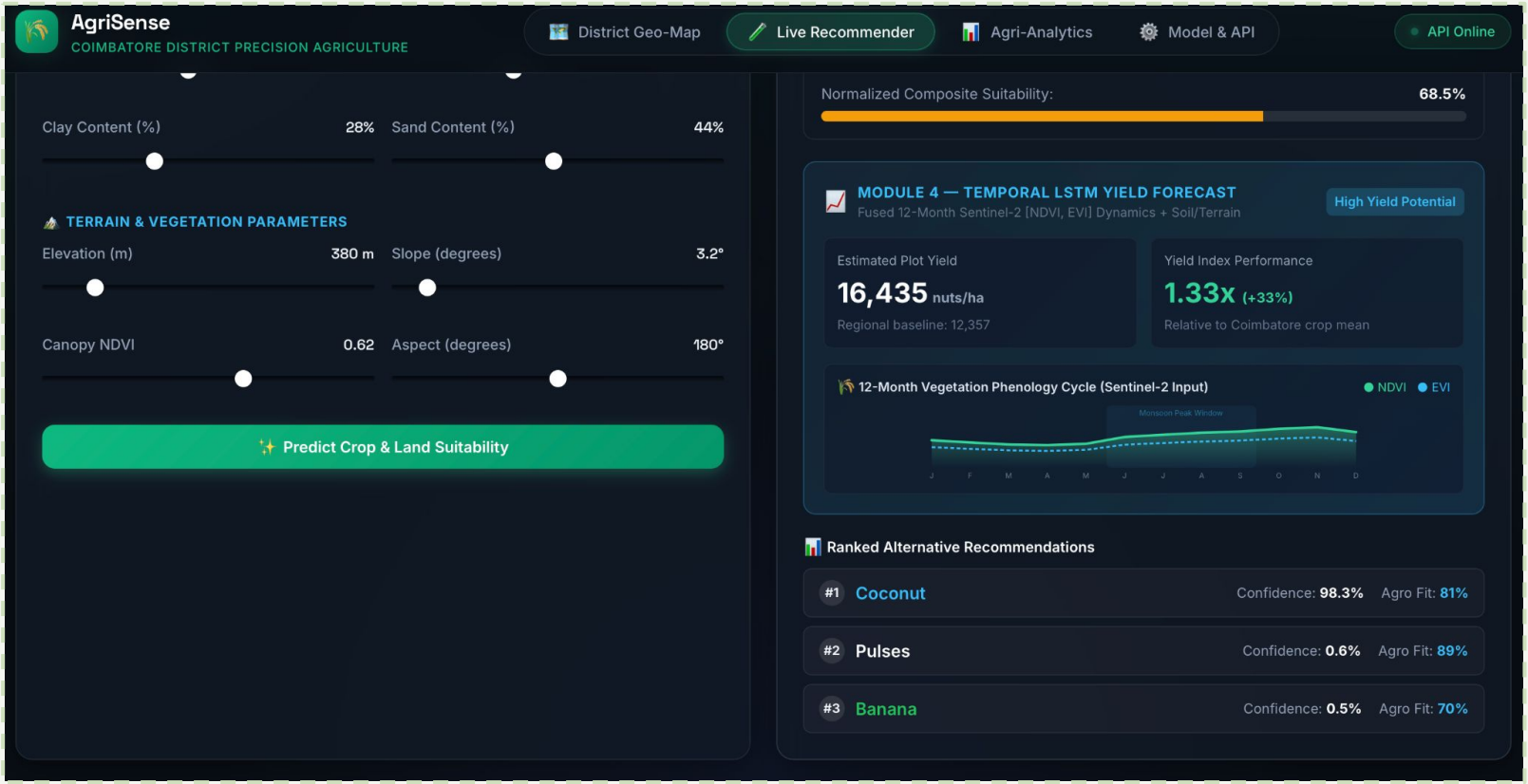
Implementation

Step 3- Crop Recommendation



Implementation

Step 4- Yield Prediction



06 · PLANNING

Timeline

Wk 1–3

Foundation

- Problem statement & literature survey
- Abstract & tech stack — Zeroth Review
- Module planning & system architecture (HLD/LLD)

Wk 4–6

Data Pipeline

- Module 1 — Data ingestion & preprocessing
- SoilGrids, SRTM, Sentinel-2/Landsat fusion
- First Review Submission

Wk 7–9

Modeling

- Module 2 — Suitability model (RF / XGBoost)
- Module 3 — Crop recommendation model
- Module 4 — Yield prediction (LSTM on NDVI/EVI)
- Second Review Submission

Wk 10–12

Explainability & Delivery

- Module 5 — Explainability engine (SHAP)
- Module 6 — Dashboard development
- System integration, testing & optimization

Wk 13–14

Evaluation & Close-out

- Evaluation on TN districts vs. reported statistics
- Final project submission & external review